

Documentation Article

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Statistical Model

Mathematical Model

Matrix Form:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

Where:

$$\mathbf{y} = \begin{pmatrix} \text{Salary}_1 \\ \text{Salary}_2 \\ \vdots \\ \text{Salary}_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

The design matrix $\mathbf{X}$ is an $n \times 5$ matrix (assuming 3 major categories with one used as the reference):

$$\mathbf{X} = \begin{pmatrix} 1 & \text{Sex}_1 & \text{Major}_{12} & \text{Major}_{13} & \text{GPA}_1 \\ 1 & \text{Sex}_2 & \text{Major}_{22} & \text{Major}_{23} & \text{GPA}_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \text{Sex}_n & \text{Major}_{n2} & \text{Major}_{n3} & \text{GPA}_n \end{pmatrix}$$

where each column represents:

- **Column 1:** Intercept (all 1s)
- **Column 2:** Sex_i (e.g., 0 = Female, 1 = Male)
- **Columns 3–4:** Dummy variables for MajorCategory, where $\text{Major}_{ij} = 1$ if observation i belongs to major j , and 0 otherwise (Major 1 serves as the reference category)
- **Column 5:** GPA_i

Prediction Justification

We evaluated our model using K-fold cross-validation with RMSE as the performance metric. The results show that the model predicts salary reasonably well relative to the overall variability in salaries in the dataset, and the RMSE values were stable across folds, indicating consistent predictive performance.

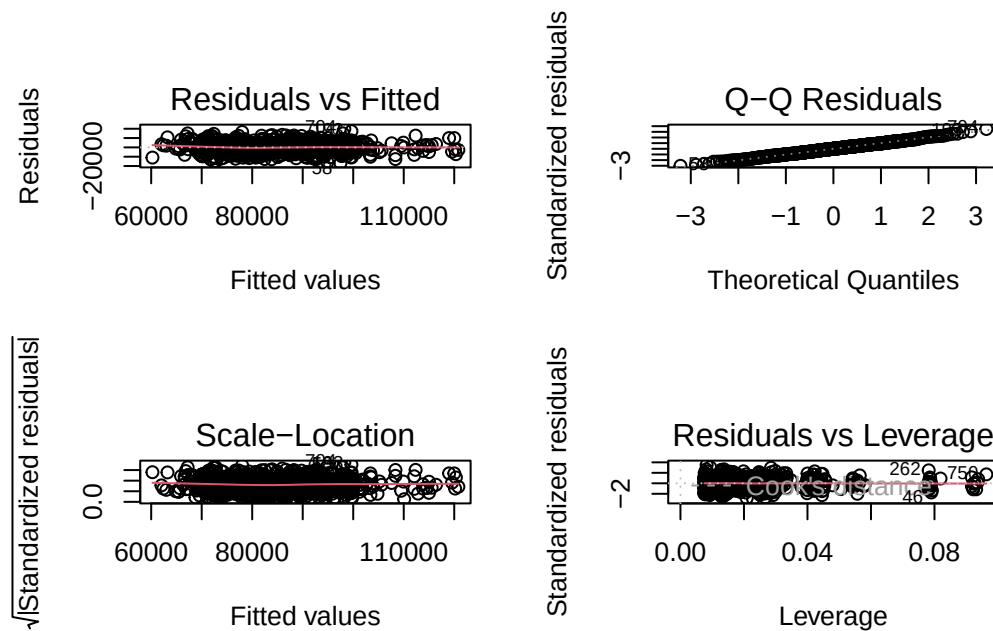
```
library(tidyverse)

sal_df <- vroom::vroom("Salary.csv")

sal_df <- sal_df |>
  mutate(
    MajorCategory = factor(MajorCategory),
    Sex = factor(Sex)
  )

# creating a linear model. Salary predicted my Major, GPA and sex. No interaction
sal_lm<-lm(Salary~ .,sal_df)

par(mfrow = c(2,2))
plot(sal_lm)
```



We also have found that our model meets all assumptions for a linear regression model. The relationship is linear, the residuals are normally distributed and homoscedastic. There is also independence across observations as far as we can tell.

Section 2

We first fit a simple linear regression of Salary on GPA (`sal_lm`). From the model summary, we extracted the estimated GPA slope ($\hat{\beta}$) and its two-sided t-test p-value to test the hypothesis that the GPA coefficient equals zero. To quantify uncertainty, we computed the 95% Wald confidence interval for the GPA coefficient using `confint(sal_lm, "GPA", level = 0.95)` and reported the estimate, p-value, and confidence interval.

To investigate whether the effect of GPA varied by Major, we introduced a GPA $\times$ Major interaction term. Because interactions require adjusted main-effect tests, we used sum-to-zero contrasts (`options(contrasts = c("contr.sum", "contr.poly"))`) and conducted a Type III ANOVA with `car::Anova(..., type = 3)` on the interaction model (`sal_lm_int_GPA_MAJ`). Interpretation combines the simple model's marginal GPA effect with the Type III ANOVA results to determine whether the GPA slope differs meaningfully across majors.

Section 3

Next, we fit a linear regression model using Major as the sole predictor of salary. The F-statistic for this model indicated that Major is a statistically significant predictor of salary. We then examined the estimated coefficients for each Major category to identify which majors were associated with higher average salaries.

Section 4

We extended our analysis by fitting a model of the form `Salary ~ GPA + Sex * MajorCategory`, which adjusts for GPA while allowing Sex $\times$ MajorCategory interactions.

Using `summary(sal_lm_sex_major)`, we evaluated the significance of the main effects and interaction. To interpret these effects, we created a balanced prediction grid containing all combinations of MajorCategory and Sex (Male and Female) at a fixed GPA of 3.5. After ensuring factor levels matched those of the training data, we obtained predicted salaries with `predict()`.

We then reshaped the predictions to compute the predicted gender salary gap for each major:

```
Delta = Salary(Male) - Salary(Female)
```

Finally, we visualized these gaps using a horizontal bar chart, ordering majors by the magnitude of the difference and coloring bars by direction ("Male > Female" vs. "Female > Male").

Section 5

To compare model complexity and predictive performance, we evaluated models with and without interaction terms. Although the F-test suggested that the interaction model was significant, its adjusted R^2 was nearly identical to the simpler model. K-fold CV also showed that the predictive RMSEs were similar. Given the minimal improvement in predictive accuracy, we prefer the simpler model.

We also compared the models' predictive RMSE to the standard deviation of salaries in the dataset. Because the RMSE was much smaller than the standard deviation, the model explains a meaningful portion of salary variability.

Code Appendix

```
#library for data cleaning and graphs
library(tidyverse)

# reading in data and creating factors
sal_df <- vroom::vroom("Salary.csv")

sal_df <- sal_df |>
  mutate(
    MajorCategory = factor(MajorCategory),
    Sex = factor(Sex)
  )

# creating a linear model. Salary predicted my Major, GPA and sex. No interaction
sal_lm<-lm(Salary~ .,sal_df)
#linear model that includes an interaction between GPA and MajorCategory
sal_lm_int_GPA_MAJ <- lm(Salary ~ GPA * MajorCategory + Sex, data = sal_df)

head(sal_df)

# 1) Pull GPA beta (estimate) and p-value from the summary object
smry <- summary(sal_lm)
coef_mat <- smry$coefficients

gpa_estimate <- coef_mat["GPA", "Estimate"]
gpa_pvalue <- coef_mat["GPA", "Pr(>|t|)"]

# 2) Confidence interval (default 95%)
gpa_ci <- confint(sal_lm, "GPA", level = 0.95)
```

```

# Print nicely
cat("GPA beta (Estimate):", round(gpa_estimate, 4), "\n")
cat("GPA p-value      :", signif(gpa_pvalue, 4), "\n")
cat("GPA 95% CI       : [",
    round(gpa_ci[1], 4), ", ", round(gpa_ci[2], 4), "]\n", sep = "")

# Recommended contrasts for Type III tests
options(contrasts = c("contr.sum", "contr.poly"))
library(car)
car::Anova(sal_lm_int_GPA_MAJ, type = 3)

# code to perform f-test between base lm, and an lm that doesn't include Major
print(anova(sal_lm, lm(Salary~ -MajorCategory,sal_df)))

# we see that major is significant,
# so now we view the summary of the model with major.
summary(sal_lm)

maj_df <- expand.grid(Sex = "F", GPA = 3.5,
                     MajorCategory = unique(sal_df$MajorCategory))

maj_preds <- predict(sal_lm, new_data = as.data.frame(maj_df))

# make df for diff majors holding constant of female w/ 3.5 GPA
maj_df <- expand.grid(Sex = "F", GPA = 3.5,
                     MajorCategory = unique(sal_df$MajorCategory))

# predict salaries for diff majors
maj_preds <- predict(sal_lm, newdata = maj_df)

# Ensure you used the correct arg for base R lm
maj_preds <- predict(sal_lm, newdata = maj_df, se.fit = TRUE)

# Bind predictions back to maj_df
plot_df <- maj_df %>%
  mutate(
    .pred = maj_preds$fit,
  ) %>%
  # Order majors by predicted value for a nicer chart
  mutate(MajorCategory = reorder(MajorCategory, .pred))

library(scales)

```

```

ggplot(plot_df, aes(x = MajorCategory, y = .pred)) +
  geom_col(fill = "#4472C4") +
  coord_flip() + # if you prefer horizontal bars (often easier to read)
  scale_y_continuous(
    breaks = scales::breaks_pretty(n = 5),
    minor_breaks = waiver(),
    labels = label_number(big.mark = ",")
  ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor.y = element_line(color = "grey85"),
    panel.grid.major.y = element_line(color = "grey80")
  ) +
  labs(x = "Major category",
       y = "Predicted Salary",
       title = "Predicted Salary by Major")

sal_lm_sex_major <- lm(
  Salary ~ GPA + Sex * MajorCategory,
  data = sal_df
)

summary(sal_lm_sex_major)

# linear model using all predictors and no interactions.
# As seen above, just defined again for clarity
sal_lm<-lm(Salary~ .,sal_df)

# linear model using all predictors and interactions found to be significant:
sal_interaction_lm <- lm(Salary ~ MajorCategory * Sex + MajorCategory * GPA,
  data = sal_df)

# function to perform k-fold cv for a single fold.
# take input of k, the kth fold
# calculates rmse for the models with and without interactions
# returns difference of rmse of the 2
cross_validate <- function(k){
  val_set <- sal_df |> filter(folds == k)
  train_set <- sal_df |> filter(folds != k)

  preds_1 <- predict.lm(lm(Salary ~ MajorCategory * Sex + MajorCategory * GPA,
    data = train_set),

```

```

        newdata = val_set)
rmse_1 <- sqrt(mean((preds_1 - val_set[["Salary"]])^2))

preds_2 <- predict.lm(lm(Salary ~ MajorCategory + Sex + GPA, data = train_set),
                      newdata = val_set)
rmse_2 <- sqrt(mean((preds_2 - val_set[["Salary"]])^2))

rmse_1 - rmse_2
}

K <- 20

# randomly creates partitions for folds
folds <- rep(1:K, length = nrow(sal_df))|>
  sample()

rmse_results <- sapply(1:K, FUN = cross_validate)

# results of k-fold validation comparison
print(summary(rmse_results))
hist(rmse_results)

# function similar to above
#instead of finding difference, just calculates rmse for 1 and returns it
cross_validate_one <- function(k){
  val_set <- sal_df |> filter(folds == k)
  train_set <- sal_df |> filter(folds != k)

  preds <- predict.lm(lm(Salary ~ Sex + MajorCategory + GPA, data = train_set),
                      newdata = val_set)
  sqrt(mean((preds - val_set[["Salary"]])^2))
}

K <- 20

# see above cell
folds <- rep(1:K, length = nrow(sal_df))|>
  sample()

rmse_results <- sapply(1:K, FUN = cross_validate_one)

# statistics from k_folds

```

```

print(sd(sal_df[["Salary"]]))
print(mean(rmse_results))
print(summary(rmse_results))

set.seed(545)

# randomly selecting a person from the df
pick <- sample(1:length(sal_df$Salary),1)

# gets explanatory variables for that person
info <- as.data.frame(sal_df)[pick,2:4]

#makes prediction for salary
prediction <- predict(sal_lm, info)

#prints explanatory variables, and also difference between prediciton and actual salary.
print(info)
print(prediction - sal_df$Salary[pick])

# creates a plot of various lm assumptions
par(mfrow = c(2,2))
plot(sal_lm)

```